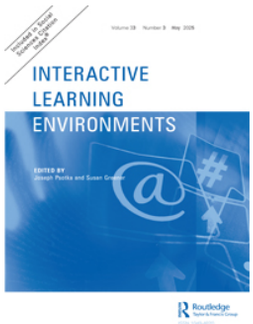

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Influence of artificial intelligence tools on higher order thinking skills: a meta-analysis

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ABSTRACT
The use of artificial intelligence (AI) has received scholarly attention and is becoming a common trend in the field of education, while the cultivation of higher order thinking (HOT) skills still remains as one of the crucial educational objectives. Recent years have seen substantial amount of research exploring the influence of Artificial Intelligence in Education (AIEd) use on each HOT skills of students, which necessitated a systematic review to amalgamate the findings quantitatively. This meta-analysis study set out to explore the effect of AIEd on students' separate HOT skills and identify the effect of potential moderator variables. The analysis included 29 articles with a total of 2.966 participants between the years of 2000 and 2023. This research confirmed the use of AIEd had a medium positive effect on skills classified as HOT skills. The findings suggested that instructional settings which incorporated AI technologies enhanced students' HOT skills 9.3% higher. This review also found that the variables of country, year of publication and AI tool/agent moderated the magnitude of effect sizes significantly. The findings offer extensive examination of AI technologies in educational settings and contribute to understanding of how and to what extent they influence students' HOT skills.

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Artificial intelligence; artificial intelligence in education; higher-order thinking skills; meta-analysis; sub-group analysis

Introduction

Today's educational systems are challenged to provide students with a variety of competencies and abilities, commonly referred to as twenty-first-century skills, in order to meet the demands of rapidly changing work environments (Luckin et al., 2016; Van Laar et al., 2017) and help students actively participate in growing knowledge societies. Among those twenty-first-century skills, cultivation of students' higher order thinking (HOT) skills has been recognised as an important educational objective (Huang et al., 2022). The acquisition of HOT skills has been acknowledged more crucial than ever in the future era of artificial intelligence (AI) (Channa et al., 2021). With the advancements in computer sciences and computational technologies, AI technologies that are autonomous, adaptive, and efficient have found widespread use in a variety of areas, such as mental healthcare (e.g. Nie et al., 2012), environmental issues (e.g. Al-Sharafi et al., 2023), engineering (e.g. Nasiri & Khosravani, 2021), and education (e.g. Wartman & Combs, 2018). In the field of mental healthcare, Nie et al. (2012) conducted a study where they utilised a supervised machine learning technique to detect and categorise mental disorders. This was accomplished by creating a model with the symptom checklist-90-R tool. Regarding another instance of AI implementation in addressing environmental issues, Al-Sharafi et al. (2023) conducted a study where they constructed a theoretical model

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using data gathered from 562 participants in Malaysia and Türkiye. The aim was to elucidate the factors that influence Generation Z's utilisation of AI products and how this usage affects environmental sustainability.

Artificial Intelligence in Education (AIEd) is one of those disciplines that has received considerable interest (Holmes et al., 2019; Ouyang et al., 2022). So far, there has been a growing body of literature that examines the effects of AIEd-driven tools/agents on students' HOT skills development and the precise effect of AIEd-driven tools is an ongoing debate. However, the systematic understanding of the overall effectiveness of AIEd-based tools on these skills remains a major challenge. In this regard, the purpose of the current study is twofold; first, to explore the influence of AIEd tools on students' HOT skills and second, to investigate the effects of several moderator variables on the effectiveness of AI technologies. To the best of our knowledge, no meta-analysis has examined the overall effect of AIEd tools on each HOT of students. Thus, this study aims not only to close the gap and provide a comprehensive analysis of the impacts of AIEd tools on several skills as HOT, but also to suggest potential relationships between variables and learning outcomes.

Literature review

Higher order thinking (HOT) skills

In the field of education, various taxonomies presented in the past have been developed to classify cognitive processes, with many sharing similarities at their highest levels. For instance, Marzano's (2001) taxonomy includes problem solving. Similarly, Haladayna's taxonomy (1997) underscores the importance of problem-solving, critical thinking, and creativity. Additionally, various scholars (Miri et al., 2007; Stein et al., 2013; Tajudin & Chinnappan, 2016; Zohar, 2013) have put forth their perspectives on the highest cognitive levels. These top cognitive levels often align with the increasingly important twenty-first-century skills framework in today's world, particularly emphasising problem-solving, critical thinking, and creative thinking.

A review of the literature reveals that various terminologies and classifications are used to define twenty-first-century skills (e.g. Binkley et al., 2012; P21, 2009 Trilling & Fadel, 2009) and HOT skills (Anderson & Krathwohl, 2001; Marshall & Horton, 2011; Miri et al., 2007). Despite these different definitions and categories, it is evident that they share similar associations and components (Wang et al., 2019) and most agree that twenty-first-century skills should encompass problem-solving, critical thinking, communication, collaboration, and creativity.

The common skills placed at the highest cognitive levels emphasised in these taxonomies and classifications are considered favourable for learning, life, career success, and social development (Tanujaya et al., 2017). Therefore, the associated skills in the context of twenty-first-century skills, as presented in a framework by the US-based Partnership for twenty-first-century Skills (P21), are creative thinking, critical thinking, and problem-solving (Kylonen, 2012; P21, 2015; Trilling & Fadel, 2009). In the present study, the selection of these skills is based on the twenty-first-century skills framework, which encompasses creative thinking, critical thinking, and problem-solving.

While there is constant discussion over the definition of critical thinking and the essential skills required for critical thinking, one particular definition and set of skills emerges as a widely accepted agreement in its conceptualisation. The Delphi Report, a report summarising the findings of this meeting, provided a definition of critical thinking as follows: "Purposeful, self-regulatory judgement which results in interpretation, analysis, evaluation, and inference, as well as an explanation of the evidential, conceptual, methodological or contextual considerations upon which that judgement is based" (Facione, 1990, p. 3). Following its adoption by the American Philosophical Association, the conceptualisation of critical thinking supplied by the Delphi Report came to be recognised as the standard for acceptable definition of critical thinking (Beckie et al., 2001; Sorensen & Yankech, 2008). In addition, the Delphi panel reached a strong consensus that analysis, evaluation, and inference were the fundamental skills required for critical thinking (Facione, 1990). In addition to its

definition, critical thinking skills are considered crucial in educational environments as they enable individuals to surpass mere knowledge retention and instead acquire a more intricate comprehension of the supplied material (Dwyer et al., 2012; Halpern, 2003). Competency in critical thinking is also crucial in social and interpersonal situations, as it facilitates effective decision-making and problem-solving in everyday scenarios (Ku, 2009). Studies indicate that individuals who possess strong critical thinking skills tend to make more effective decisions and judgments when faced with complex situations (Gambrill, 2006). They are also less prone to cognitive bias and heuristic thinking (Facione & Facione, 2001; McGuinness, 2013). Furthermore, these individuals are more likely to achieve higher grades, acquire greater knowledge, actively participate in society, and have increased employability (Barton & McCully, 2007; Holmes & Clizbe, 1997; National Academy of Sciences, 2005).

Creativity can be conceptualised as the process of bringing things together in new ways (conceptually or artistically), building something new, and working to create an interesting situation using unusual or uncommon circumstances (Brookhart, 2010). Students with creative skills should possess a wide range of qualities, such as using idea generation techniques (like brainstorming), generating new and valuable ideas, developing creative ideas, and thoroughly examining, analysing, and evaluating their own ideas to maximise their potential (P21, 2015; Trilling & Fadel, 2009).

Problem-solving can be defined as the mental steps required for achieving a goal or overcoming a challenge. Students with problem-solving skills should be able to solve unknown problems of various types through both traditional and innovative methods, explain different perspectives, and possess the ability to ask and define significant questions to find better solutions (Haladyna, 1997).

With the growing significance of twenty-first-century competencies, numerous institutions have intensified their efforts to define these HOT skills (e.g. ISTE, 2016; P21, 2009). These institutions indicate that being successful in the twenty-first-century requires possessing multiple competencies. The first reason regarding the importance of these skills is economical. More precisely, computers and robots are capable of performing activities that individuals with minimal knowledge and skills can perform. As a result, the labour force requires fewer individuals with simple fundamental skill sets and more individuals with several HOT skills. In addition, supply and demand in a global market as opposed to a national or local market creates rivalry amongst people who can create something new by using non-routine, complex thinking, and communication skills and apply them to new issues and circumstances (Saavedra & Opfer, 2012). Moreover, as a second reason for the importance of developing each HOT skill, schools have a compelling civic incentive to place greater emphasis on equipping students with each HOT skill. Despite the fact that students must have a foundation of fundamental civic knowledge, rote learning and reciting facts about governance and citizenship are insufficient for fostering civic involvement. Additionally, students must be taught the how and why of critical thinking in order to be active citizens who can, for instance, assess news cases and determine biases. They must be capable of solving problems in order to suggest or evaluate policies to address social issues. The third reason for developing separate HOT skills is related to globalisation (Boix-Mansilla & Jackson, 2011). Massive global migration, the Internet, lengthy flights, interdependent international markets, climate instability, international conflicts, and other factors serve as daily reminders that people are a part of the human race and that nations, states, and individuals are all connected to one another politically, economically, and ecologically.

In light of this complicated interdependence, the necessity for the cultivation of critical thinking skills becomes apparent, as students are required to effectively negotiate complicated global issues and arrive at well-informed conclusions that take into account a range of views and global implications. The development of critical thinking skills empowers students to engage in the analysis of intricate scenarios, assess the validity of evidence, and ultimately reach well-founded judgments, preparing them to handle the intricate web of global issues effectively (Appiah, 2006). Furthermore, the interconnected nature of the world calls for students to enhance their ability to collaborate and

work with others from different cultural backgrounds, nationalities, and perspectives. This cross-border collaboration is vital in solving global problems that require collective efforts and diverse expertise. By collaborating with individuals from different parts of the world, students can gain a deeper understanding of global challenges, draw on diverse ideas and solutions, and foster a sense of global citizenship (Osler & Vincent, 2002). In this context, new pedagogical methods, content, teaching technologies, and strategies for learning and assessment need to be developed in educational environments (Gordon et al., 2009) which should be designed to encourage students to be more active, think critically, solve problems, and collaborate with each other (O'Neal et al., 2017).

The use of AIEd tools/agents

AIEd places an emphasis on using AI to support instructional practices, facilitate students' learning processes, enhance model-based reasoning, which includes analysing cause and effect associations, various HOT skills encompassing critical thinking, problem-solving, and making inferences (Rus et al., 2013), and advance the transformation of educational systems (Chen et al., 2020; Holmes et al., 2019; Hwang et al., 2020). Particularly, the functions of AIEd tools can be divided into three categories. Firstly, a number of studies (Mikropoulos & Natsis, 2011; Rus et al., 2013; Sharma et al., 2019) have revealed that AIEd tools have a role in several administrative tasks including students' assignments and paper reviews, grading, and providing feedback. For example, as highlighted by Rus et al. (2013) teachers who use intelligent tutoring systems (ITS) develop their effectiveness in providing students with guidance and feedback. Secondly, AIEd-driven agents plays an important role in enhancing the instructional processes. More precisely, AI has facilitated the development of various pedagogical tools, platforms and technologies that improve instructional quality. As reported in several other studies (e.g. Mikropoulos & Natsis, 2011; Wartman & Combs, 2018), virtual reality, 3-D technology and interactive simulation are examples of technologies to present or show students concepts or exhibit materials and to provide students with experiential or practical learning opportunities. The researchers further have added that these tools allow students to build a deeper understanding by allowing them to demonstrate the concepts. In addition to the previously described tools and technologies, chatbots and computer-based robots with dialogue capabilities can be utilised to respond to students' common inquiries (Pokrivcakova, 2019). As an additional application of AIEd, a considerable amount of studies (Kahraman et al., 2010); Peredo et al., 2011; Pokrivcakova, 2019; Rus et al., 2013) has examined the use of intelligent teaching systems in various forms. Peredo et al. (2011), for instance, explains the use of intelligent and adaptable web-based systems, in which teachers are observed and presented as social agents. The system then seeks to comprehend and assist teachers in carrying out their responsibilities, as well as providing students instructional guidance, with the goal of ensuring that the web-based technology used in education is an effective and systematic means of enhancing learners' experiences. Lastly, learning is another component of education where AIEd has been incorporated and deployed. Particularly, much research (Chassignol et al., 2018; Mikropoulos & Natsis, 2011; Sharma et al., 2019) has been conducted on the function of AIEd tools in individualising the curriculum and content in accordance with students' needs and abilities, hence enhancing their learning. These research further illustrate that using machine learning algorithms, various applications or platforms such as Knewton, chatbots (Pokrivcakova, 2019), Cerego and CALL (Chassignol et al., 2018) can infer a user's preferred learning approach and then tailor the course content to that person's preferences in real time. Furthermore, AIEd tools allow access to global learning through online and web-based platforms, lowering some of the barriers to learning possibilities at national and international level (Sharma et al., 2019). Data from other studies suggest that as a means of enhancing the learning experiences of students, simulation-based learning and ITS can be used to foster deep learning. For example, in their study, Kahraman et al. (2010)) indicate that a component of Adaptive and intelligent Web-based educational systems provides intelligent assistance to students at each stage of the problem-solving process. Peredo et al.

(2011), like Kahraman et al., argue that intelligent web based systems, particularly multi-agent systems, treat the students as social agents, examine their behaviours and adapt instruction individually. As a result, AIEd-based tools has been shown to play an important role in the teaching and learning process and have an impact on individual experiences.

The impact of AIEd tools/agents on HOT skills

Extensive research has been published on the diverse impacts of various branches and tools of AIEd on several higher-order thinking skills. However, there is a lack of consistency in their findings. While some studies indicate a positive impact of AIEd tools, some others examining the effect on the same higher order thinking skill report either no significant or a negative influence.

As for the positive effect of AIEd tools on critical thinking, it's worth noting that simulations and games also contribute to the enhancement of it (Spector & Ma, 2019). These technologies immerse students in interactive learning experiences, challenge them to apply their knowledge to real-world problems and develop skills like analysis, synthesis, and evaluation (Almeida & Simoes, 2019). Furthermore, AI-powered natural language processing tools facilitate meaningful discussions between students and virtual assistants or Chatbots by promoting critical thinking through contextually relevant responses (Barsoun et al., 2022; Cotton et al., 2023; Goda et al., 2014; Hapsari & Wu, 2022). For example, Barsoun et al. (2022) who investigated the effectiveness of using a cognitive style based Chatbot in developing science concepts and critical thinking skills among eighth graders reported that using the cognitive style-based chatbot has a positive effect in development of critical thinking. In a similar vein, Goda et al. (2014) conducted an experimental research to investigate EFL students' critical thinking skills after their preceding conversation with Chatbot. They found that pre-discussion with a Chatbot increased the students' awareness of critical thinking and enabled them to form inquiring mindsets. Moreover, existing research has postulated the supporting role of AI technologies on the development of creativity-related skills like curiosity, grit, persistence, and attentiveness (Belpaeme et al., 2018; Gordon et al., 2015). Ryu and Han (2018) discovered that experienced teachers perceive AI as a means to improve creativity. However, while some findings indicate significant contributions of various AI tools, such as Deep Learning recommendation Based augmented reality System and Chatbot, to the development of creativity (Diamond, 2013; Rudolph et al., 2023) and critical thinking (Barsoun et al., 2022; Goda et al., 2014; Hapsari & Wu, 2022; Lin & Chen, 2020; Soliman, 2019), some others provide negative accounts about AI use, by arguing that, for example, relying heavily on the content generated by Chatbot and ChatGPT can overshadow these skills in students (Kooli, 2023).

A similar situation applies to the impact of AIEd tools on problem solving skills. Kafai and Burke (2014) highlight that AIEd aims to encourage problem-solving through collaboration and going beyond mere knowledge acquisition. Similarly, in their single-case study analysing interactions between students and AI tools, such as ChatGPT and Bing Chat, in simulated STEM learning experiences, Vasconcelos and dos Santos (2023) point to the role of AI tools in the development of problem-solving skills, critical thinking skills and creativity. Within the branch of Intelligent Tutoring Systems (ITS), artificial intelligence algorithms play the role of virtual teachers by giving each student specific feedback and instructions. Fostering problem solving skills through interactive learning experiences, these systems can evaluate a student's understanding and adjust the educational strategy accordingly (Jeremic et al., 2012; Nabiyev et al., 2013). For instance, Hooshyar et al. (2016), who evaluated the effect of a Flowchart-based Intelligent Tutoring System (FITS) on university students' problem-solving skills, discovered that students' problem-solving skills improved more when they were exposed to FITS. However, several studies (Beal et al., 2010; Kooli, 2023; Maniktala et al., 2023) report that employing chatbots or any other AI-based tools has the potential to seriously affect students' academic advancement and their capacity to acquire knowledge and that they create a dependency and hinder independent problem-solving abilities. Along the same lines, the Beal et al. (2010) study which investigated the effect of ITS on middle school students' problem-

solving skills, reported higher post-test performance for students who did not use ITS. In a similar research by Maniktala et al. (2023), higher skills were attained by non-ITS university students.

Overall, the inconsistent findings in related literature highlight the need to consolidate existing research results in order to gain a better understanding of how AIEd tools affects various HOT skills. Thus, a rigorous methodology, such as a meta-analysis, with a capacity of synthesising research findings and eliminating bias (Kates et al., 2018) is required to estimate the overall effect size across multiple investigations (Borenstein et al., 2009). By employing meta-analysis, this study is believed to provide a systematic understanding of how or to what extent AIEd tools contributes to the development of students' HOT skills.

Related review and meta-analyses of AIEd

To date, existing systematic reviews and meta-analyses of AIEd have primarily focused on AI's effectiveness on different outcomes such as learning (e.g. Tan et al., 2022; Zheng et al., 2023), perception (e.g. Zheng et al., 2023), and achievement (e.g. Cruz-Jesus et al., 2020; Hwang, 2022; Zheng et al., 2023), but none exists for its effectiveness on separate HOT skills. Zheng et al. (2023) conducted a meta-analysis of 20 years of study on the influence of AI technology on learning achievements and perceptions of people. Similarly, Hwang (2022) used meta-analysis to examine the impact of AI agents on primary school students' mathematics achievement. Consequently, past research summaries dealt with students' skills without distinguishing their cognitive levels. There is a need to differentiate HOT skills from other outcomes or skills and undertake a separate systematic review of the effectiveness of AI technology on them.

Moreover, various studies synthesised research results on effectiveness of AIEd agents in higher education (e.g. Zawacki-Richter, 2019), but there remains a paucity of evidence on its effectiveness in other educational levels (e.g. García-Martínez et al., 2023). Additionally, in a recent meta-analysis study, García-Martínez et al. (2023) investigated the impact of AI components and computational sciences on student performance. However, the generalisability of that research is limited since it only included studies published between 2010 and 2020. In other words, studies in the first decade of the twenty-first-century, and those completed in the last three years reflecting the current situation were not included.

The present study

Having presented the inconsistent results in literature, the current meta-analysis set out to fill the gap and offer a thorough examination of research results published between 2000 and 2023 to explore influence of AIEd tools on various HOT skills. We believe this study has a potential of significant contribution, because not only it aims to provide an updated review of extant research on AIEd tools' influence on students' HOT skills, but also it offers moderator analyses to investigate the associations between possible factors and learning outcomes (Chen et al., 2018). Thus, considering the call of scholars (Zheng et al., 2023), this study performed a meta-analysis to address the following research questions:

(1) What is the overall influence of AIEd tools on students' higher-order thinking skills?
(2) How do several moderator variables (year, sample, education level, higher-order thinking skill, AI branch, and AI agent) influence the effects of AIEd tools?

Methodology

Meta-analysis is defined as a set of statistical techniques for synthesising the results of a body of studies (Borenstein et al., 2021). By summarising the results of studies, meta-analyses help researchers and

decision makers give evidence-based policy decisions (Pigott & Polanin, 2020). Given its significance, this meta-analytic study used comprehensive methodology to promote both transparency and reproducibility as much as possible (Pigott & Polanin, 2020; Stewart et al., 2012). For research purposes, the recommendations of the recently revised PRISMA statement (Page et al., 2021) were used in reporting. In short, the six steps suggested by Field and Gillett (2010) were followed: (1) completing the search of the literature, (2) determining and applying the inclusion and exclusion criteria for study selection, (3) calculating the effect sizes, (4) completing the basic statistics, (5) conducting complex analyses (e.g. publication bias, sub-group analyses), and (6) reporting the results.

Literature search and inclusion criteria

In line with the rationale as explained in the introduction, the related research which explored the effects of AIEd tools on HOT skills was the main focus of this systematic review. A search of the Web of Science, Scopus and Educational Resources Information Center (ERIC) was conducted to locate the studies using the following keywords illustrated in Table 1. The reason to locate studies in WOS, Scopus and ERIC was due to the huge size of these databases, which include a broad range of good quality research. Also, the abundance of studies overlapping in other databases in Rayyan (https://rayyan.ai/), a web-based text mining application, also serves as evidence for the comprehensiveness of these databases in the field of education. The keywords were identified after a thorough review of extant research on AIEd and HOT skills (Ouyang et al., 2022; Saepuzaman et al., 2021; Zheng et al., 2023). Additionally, reference lists of several studies were reviewed and researchers were requested to send any unpublished data. Since twenty-first-century skills, as the name denotes, received emphasis at the early years of 2000s, the search limit in years was identified as 2000–2023, in which the search ended in April 2023.

Non-refereed reports, books, magazine articles, essays and nonempirical studies were excluded in the initial search. Overlapping sources and duplicates were detected with Rayyan (https://rayyan.ai/) and subsequently removed from the list. All these efforts yielded 1.902 potential studies (from 2.440 in total), which were screened for inclusion by taking the following criteria into account:

- Studies were included if they were true or quasi-experimental which explored the effect of AI technologies on HOT skills.

Table 1. Keywords used in search of related studies.

AIEd-Related Terms	HOT skills Related Terms
"AI"	"critical thinking skill*"
"AIEd"	"problem solving skill*"
"artificial intelligence"	"problem-solving"
"artificial intelligence in education"	"higher order thinking"
"machine intelligence"	"high-level skill*"
"intelligent support"	"creativity"
"intelligent virtual reality"	"creative thinking"
"chat bot*"	
"machine learning"	
"automated tutor*"	
"personal tutor*"	
"intelligent agent*"	
"expert system*"	
"neural network*"	
"learning analytic*"	
"deep learning"	
"natural language processing"	
"chatbot*"	
"intelligent system"	
"intelligent tutor*"	

- Studies were included if they were completed in the field of education.
- Studies were included if they provided usable data for calculating the effect size.

Based on these criteria, 29 studies were included, from which 37 effect sizes ($N = 2.966$) were extracted. Several studies (e.g. VanLehn et al., 2005) provided more than one effect size depending on various samples present in their research design. The PRISMA flowchart (Page et al., 2021) in Figure 1 illustrates the selection process for studies in the present meta-analysis. The summary of studies included can be reviewed (Table A) in the Supplementary Document.

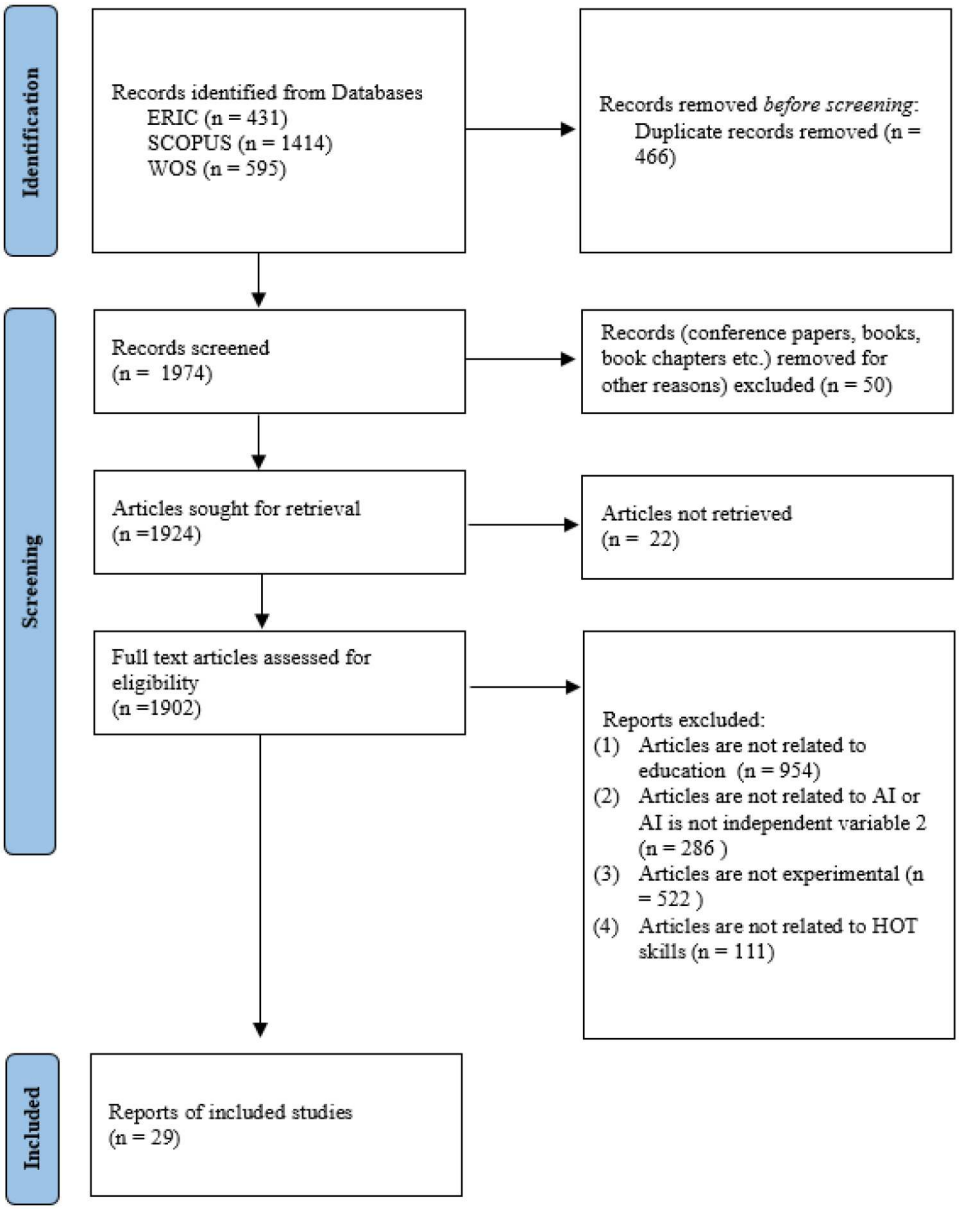

Figure 1. The PRISMA flowchart.

Data analysis

We converted Cohen's d to Hedges' g to determine the extent of the difference between HOT skills scores of experimental and control groups. Similar to Cohen's d, Hedges' g is calculated by dividing the mean difference by the pooled standard deviation. Due to the fact that the sample pooled standard deviation tends to underestimate the population variance, a correction is made in the computation, resulting in a less biased measure of effect size (Lambert & Alhassoon, 2015). In addition, some studies included in the current meta-analysis used multiple measures to reflect the same dependent variable or reported separate results for test subscales, resulting in reporting multiple, statistically dependent effect sizes. Therefore, we aggregated dependent effect sizes so that such studies (e.g. Lin & Chen, 2020) would only contribute one effect size per outcome domain in order to account for effect size dependence. In the process of aggregation, since it is considered the least biased and the most accurate, we followed the Borenstein, Hedges, Higgins, and Rothstein method that accounts for within-study correlations (Borenstein et al., 2021). We also ran sensitivity analysis to determine if the conclusions differed depending on the relationship among dependent effect sizes. Specifically, as suggested by Borenstein et al. (2021), we performed distinct aggregations and analyses at deviant values (e.g. = 0.40;= 0.60). To aggregate the effect sizes and perform sensitivity analysis, we used MAd package (Del Re & Hoyt, 2014) in R (R Core Team, 2014).

Further analyses were carried out using Comprehensive Meta-Analysis Version 4 (Borenstein et al., 2022). The effect sizes were interpreted according to Cohen's (1988) suggestion as small (0.2), medium (0.5) and large (0.8). The validity of this research was ensured by the detailed search of the literature, inclusion of eligible studies, and reporting of effect sizes. Three researchers with a doctoral degree reached full agreement in the selection of studies, and analysis and interpretation of the data, which ensured the reliability of this research. In the coding process, the researchers independently examined the included studies, shared their codings, discussed and resolved the discrepancies.

The random-effects model was used in this meta-analysis since it was assumed that AIEd's influence on several HOT skills differed beyond what might be attributable to sampling error. The assumption that the included experimental studies would have varying degrees of effect sizes due to different samples from different nations with different sizes and different AI technologies and HOT skills informed the selection of a random-effects model approach. Additionally, the random-effects model is specifically recommended for meta-analysis of studies in social sciences (Field & Gillett, 2010) and the results of random-effects model approaches are suitable for generalisations (Borenstein et al., 2021).

Apart from the mean effect size analysis, the data were tested for heterogeneity and publication bias (Rothstein et al., 2005). I2 and Q statistics were used to quantify the extent of heterogeneity among the studies included (Higgins et al., 2003; Higgins & Thompson, 2002). A statistically significant Q value (e.g. $p < .01$) would inform that the data were heterogeneous. I2, which "describes the percentage of total variation across studies that is due to heterogeneity rather than chance" (Higgins et al., 2003), may present low, moderate and high heterogeneity considering values of 25%, 50% and 75%, respectively (Higgins & Thompson, 2002). Publication bias, which is considered a significant challenge (Rothstein et al., 2005), informs about possible bias in the selection of eligible studies, resulting from likelihood of including publications with strong results (Higgins et al., 2019; Sutton et al., 2000). The funnel plot, Egger's Regression Test, Begg and Mazumdar Rank Correlation, Duval & Tweedie's Trim and Fill (TFM), and Rosenthal's FSN were employed to detect the publication bias in the present study. For this purpose, the funnel plot was inspected for asymmetry and suggestions of TFM for additional studies and adjusted mean effect size after imputation were assessed. The nonsignificant p value (>.05) in Egger's Regression Test and Begg and Mazumdar Rank Correlation would yield the nonexistence of bias. The $5k + 10$ (where k represents the number of studies included) was used as a cut-off point for Rosenthal's FSN, where a higher value than the $5k + 10$ calculation would support the non-existence of publication bias (Fragkos et al., 2014; Mullen et al., 2001).

Besides the common effect calculations, based on Lipsey and Wilson's (2001) suggestions, analogue to ANOVA, which "provides a method of testing the ability of a single categorical variable, such as treatment type, to explain excess variability in a distribution of effect sizes" (p. 122), was utilised to examine the influence of the moderators (year of publication, sample, type HOT skills, level of education, AI branch, and AI tools or agents) on mean effect size. As a Chi-Square-based sub-group analysis method (Yıldırım et al., 2019), analogue to ANOVA explains effect-size differences in subgroups. In other words, a between-groups heterogeneity (QB) value above the critical chi-square value would suggest a significant effect ($p < .05$). The selection of the publication year as a moderating factor was motivated by researchers' desire to observe the trajectory of studies concerning the intersection of AI and HOT skills since the onset of the twenty-first century. The choice of country as another moderating variable stemmed from the globally recognised push towards integrating AI technologies into educational curricula (e.g. European Commission, 2018). This choice aimed to determine whether countries exhibited variations in their utilisation of AI within education systems, and whether such disparities influenced students' higher order thinking outcomes. AI has found application within instructional and learning processes (Chen et al., 2020), offering the advantage of expanding learning opportunities, enriching learning experiences, and fostering students' cognitive skills (Zhang & Aslan, 2021). Nevertheless, prior studies have not comprehensively investigated the overall impact of AI on HOT skills. As a result, the type of HOT skills as distinct twenty-first-century thinking skills was incorporated as another moderator variable. The educational level also emerged as a significant consideration, as AI technologies hold strategic importance within education, providing effective learning environments across different levels of education (Zhai et al., 2021). Finally, to observe the potential impact of distinct AI tools/ agents on HOT skills, the classification scheme proposed by Zhang and Aslan (2021) who report a comprehensive review of articles and the current state of AIEd was used to categorise the AI applications under the title "AI branch" to see the extent of their effect on thinking skills when they are used to assist expert opinion, act as intelligent tutors or agents or provide personalised learning systems or environments. The subgroup analyses finish with effect-size differences of each AI tool or agent used in extant research.

Results

The review considered a total of 29 papers with 37 effect sizes. The effect size index was calculated by taking the bias-corrected standardised mean difference (Hedges' g). The random-effects model was used in the statistical analyses, since it was expected that the studies included in this analysis were a random selection from a larger pool of possible research, and the results of this analysis would be used to draw conclusions about the larger pool (Borenstein, 2019; Borenstein et al., 2010, 2021; Hedges & Vevea, 1998; Higgins & Thomas, 2019). The results of the mean effect size, the heterogeneity test, the publication bias statistics, and the sub-group analyses are presented in subsequent sections.

Findings of mean effect size

The mean effect size and the result of the heterogeneity test for the studies included are shown in Table 2.

The random effects model reflected the mean effect size as .63 (SE = 0.08, 95% CI = [−0.25, 1.50]). This value implied a medium positive effect (Cohen, 1988) of AIEd tools on HOT skills. The distribution of true

Table 2. Effect size and heterogeneity test result.

					95% CI					
Model	*N*	*Mean ES*	*Z*	*SE*	Low	Up	*df*	*Q*	*p*	I^2
Random	37	0.628	7.447	0.084	−0.246	1.502	36	151.509	0.000	76.239

effects as presented in Figure 2 shows that the mean effect size is .63 with a confidence interval of .46 to .79. The Q value of 151.51 ($p < .01$) informed heterogeneous data and the null hypothesis of homogeneity is rejected since the Q value "exceeds the critical value for a chi-square with k – 1 degrees of freedom" (Lipsey & Wilson, 2001, p. 116). Additionally, in accordance with Higgins and Thompson (2002), the I2 value of 76.24% indicated high heterogeneity. The forest plot (Figure 3) illustrates the effect sizes included in this random effects model. These analyses and the forest plot suggests medium significant effect of AIEd tools on HOT skills and shows that the data were highly heterogeneous.

The funnel plot, Egger's Regression Test, Begg and Mazumdar Rank Correlation, Duval & Tweedie's Trim and Fill (TFM), and Rosenthal's FSN were employed to detect publication bias. Aligned with the symmetric shape of the funnel plot, TFM imputes no additional studies to either side of the mean as illustrated in Figure 4. The respective p values of .18 and .15 for the Egger's Regression Test and the Begg and Mazumdar Rank Correlation (>.05) and the need for 1.981 additional studies (larger than cut-off point of 195) with non-significant results in Rosenthal's FSN calculations confirm the robustness of the data and reflects the non-existence of publication bias.

Sub-group analyses

In addition to the mean effect size calculations, group comparisons were performed to find out the sources of heterogeneity. Analogue to ANOVA was used to explore the moderating effect of year of publication, sample, type of thinking skills, education level, AI branch, and AI tools/ agents as the dependent variables on effect sizes. Except for education level, thinking skill and AI branch, the between-groups heterogeneity (QB) was found above the critical chi-square value for all the remaining sub-groups, which suggested a significant effect ($p < .05$). In other words, the variables of year of publication, country, and type of AI tool/ agent used were found to be moderating the magnitude of effect sizes significantly. The summary table below illustrates the sub-group analysis findings with effect sizes presented in descending order for each moderating variable. The detailed results can be reviewed in Tables B-G in the supplementary document.

Discussion and conclusion

This study aimed to determine the exact impact of AIEd tools on students' problem solving, creative thinking and critical thinking skills, as the role of AI on thinking skills is becoming increasingly important in educational settings (Hopcan et al., 2023). Meta-analysis was used to amalgamate the related research findings quantitatively. The first question sought to determine the influence of AIEd tools

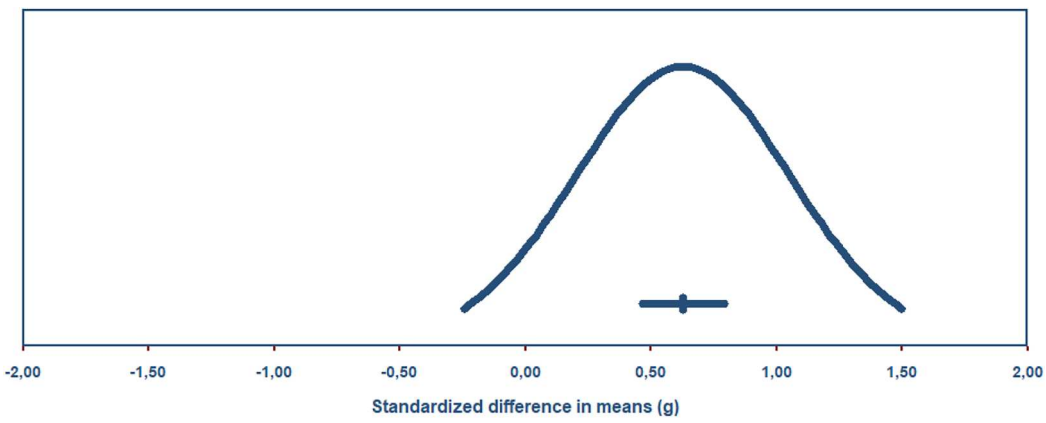

Figure 2. Distribution of true effects.

Study name	Hedges's g	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Karaoglan Yilmaz, 2022	0,872	0,311	0,096	0,263	1,481	2,807	0,005
VanLehn et al., 2005a	0,230	0,109	0,012	0,015	0,444	2,101	0,036
VanLehn et al., 2005b	0,781	0,125	0,016	0,536	1,026	6,257	0,000
VanLehn et al., 2005c	0,526	0,176	0,031	0,181	0,871	2,987	0,003
VanLehn et al., 2005d	0,449	0,173	0,030	0,110	0,789	2,593	0,010
VanLehn et al., 2005e	0,654	0,186	0,035	0,289	1,019	3,512	0,000
Arnau et al., 2013	0,535	0,240	0,058	0,065	1,006	2,229	0,026
Khodeir et al., 2017	1,426	0,400	0,160	0,642	2,211	3,565	0,000
Beal et al., 2010a	0,196	0,388	0,151	-0,565	0,956	0,504	0,614
Beal et al., 2010b	0,515	0,443	0,197	-0,354	1,384	1,162	0,245
Beal et al., 2010c	-1,034	0,436	0,190	-1,889	-0,179	-2,369	0,018
Winkler et al., 2021a	0,496	0,298	0,089	-0,087	1,079	1,667	0,096
Winkler et al., 2021b	1,713	0,344	0,118	1,039	2,388	4,978	0,000
Soliman, 2019	1,992	0,439	0,192	1,132	2,851	4,541	0,000
Rosen, 2015	-0,239	0,175	0,031	-0,581	0,103	-1,369	0,171
Hsu & Chen, 2022	0,610	0,263	0,069	0,094	1,126	2,317	0,021
Jang et al., 2022	0,776	0,133	0,018	0,514	1,037	5,811	0,000
Chu et al., 2014	0,403	0,180	0,033	0,049	0,756	2,232	0,026
Johnson et al., 2009	-0,040	0,267	0,071	-0,563	0,484	-0,149	0,882
Pol et al., 2008	1,210	0,356	0,127	0,512	1,907	3,400	0,001
Lanzilotti & Roselli, 2007	0,093	0,310	0,096	-0,515	0,701	0,300	0,764
Mitrovic, 2003	0,497	0,176	0,031	0,152	0,842	2,823	0,005
Jaques et al., 2013	0,723	0,309	0,096	0,117	1,330	2,337	0,019
Arevalillo-Herraez et al., 2013	0,959	0,336	0,113	0,300	1,618	2,852	0,004
Arnau et al., 2014	1,007	0,347	0,121	0,326	1,687	2,898	0,004
Cardenas-Robledo & Pena-Ayala, 2019	-0,291	0,452	0,204	-1,176	0,594	-0,645	0,519
Sung et al., 2018	0,735	0,280	0,078	0,186	1,284	2,626	0,009
Hooshyar et al., 2016	2,376	0,362	0,131	1,667	3,084	6,571	0,000
Lee, 2007	0,213	0,220	0,048	-0,219	0,644	0,967	0,334
Hooshyar et al., 2018	1,388	0,386	0,149	0,630	2,145	3,591	0,000
Maniktala et al., 2023	0,450	0,233	0,054	-0,007	0,907	1,930	0,054
Lane & VanLehn, 2005	0,402	0,324	0,105	-0,233	1,037	1,241	0,215
McKenna & Agogino, 2004a	0,042	0,367	0,135	-0,678	0,762	0,114	0,909
McKenna & Agogino, 2004b	-0,240	0,370	0,137	-0,965	0,485	-0,648	0,517
Xin et al., 2017	1,135	0,428	0,183	0,297	1,973	2,653	0,008
Hooshyar et al., 2015	1,639	0,360	0,130	0,933	2,345	4,551	0,000
Wang et al., 2015	0,929	0,182	0,033	0,572	1,287	5,100	0,000
Pooled	0,628	0,084	0,007	0,463	0,793	7,447	0,000
Prediction Interval	0,628			-0,246	1,502		

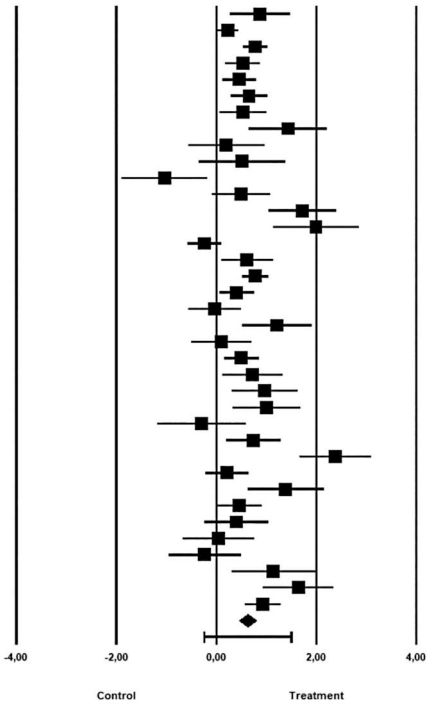

Figure 3. Forest plot showing the distribution of effect sizes.

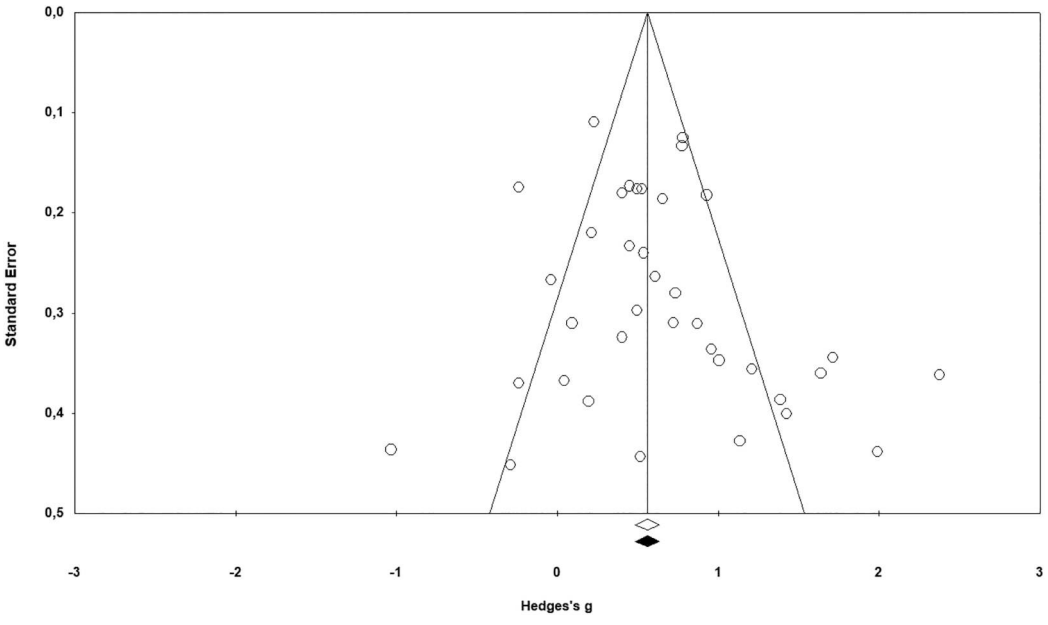

Figure 4. Funnel plot of standard error by Hedges' g.

on students' problem solving, creative thinking and critical thinking skills classified as HOT skills. The results revealed that AIEd tools had medium-level positive effect on these skills based on 29 articles published between the years of 2000 and 2023 and by aggregating participants in included studies, the total sample size reached 2966. This calculated value of mean effect size (Mean $ES = 0.628$) is equivalent to the eta-squared value of .093. In other words, the findings suggested that instructional settings which incorporated AI technologies enhanced students' problem solving, creative thinking and critical thinking skills 9.3% higher than other settings which were generally traditional. This finding shows consistency with some previous research (Arnau et al., 2013, 2014; Brown et al., 2020; Hopson et al., 2001; Wu et al., 2013), the results of which suggest that students in educational settings which utilise AI technologies attain higher levels at skills of problem solving, critical thinking and creative thinking. AI technologies have been shown to increase effectiveness of learning environments, personalise learning, guide both students and teachers through teaching and learning processes (Drigas & Ioannidou, 2012) and ensure suitable learning environments (Zheng et al., 2023). Additionally, AI technologies have been addressed to enable monitoring and evaluation of complex thinking skills, encourage different perspectives, foster learning wherever it takes place, support students with special needs, provide more accessible ways for student self-improvement, and provide learning environments in which students are allowed to interact with the instructional content (Hopcan et al., 2023; Karsenti, 2019; Luckin et al., 2016; Zhonggen & Xiaozhi, 2019). This combination of findings in related literature is suggestive of AIEd's potential in enhancing problem solving, critical thinking and creative thinking skills, which is also reflected in the results of our systematic review.

Besides the analysis of mean effect size, some moderator variables were examined to find the sources of heterogeneity. For this purpose, analogue to ANOVA was performed to investigate the moderating effect of year of publication, country, type of thinking skill, education level, AI branch, and AI tool/agent used. The subgroup analyses confirmed that year of publication, country, and AI tool/agent were sources of variance (moderating the magnitude of effect sizes), while no significant differences were suggested considering education level, type of thinking skill under focus, and AI branch. Although found nonsignificant, the positive moderator effect of education level reveals some evidence about the development of AI technologies in all educational settings. A quick review of education levels (Table 3) indicates similarity in number of studies and effect sizes through education levels. The finding about similar or close AIEd effects on HOT skills in different education levels (from elementary school to universities) shows promise in two ways. First, the interest towards the use of AI technologies is the same at all levels and second, AI technologies are incorporated to all classes and can be utilised effectively in the development of HOT skills at all education levels. This finding is also supported by the results of various systematic review studies that investigated the use of AI in both K-12 education (Zafari et al., 2022) and higher education (Zawacki-Richter et al., 2019). The results obtained from these studies also suggest that the number of research studies utilising AI is progressively increasing at both the K-12 level and higher education level, indicating the relevance of employing AI across different levels of education. However, as indicated in Table 3, the majority of the research was conducted with undergraduates. There is a requirement for an increased amount of research at K-12 level, given the numerous opportunities that the field of AI presents within those levels.

Considering the finding of significant differences among different years of publication as a moderator variable, the gradual increase in effect sizes may not only reflect the development of AI technologies in years, but also exhibit growing interest in educational settings. The highest effect sizes of the periods of 2013–2017 and 2018–2023 may show us the advances in AI technologies in recent years and also indicate better use of them in the enhancement of HOT skills. This finding resonates well with earlier study on the topic of flow, which shows a significant rise in the number of research papers focusing on AI applications in higher education published since around 2010 and beyond (Zawacki-Richter et al., 2019). Report of the Horizon also draws attention to the increasing potential

Table 3. Summary of sub-group analysis findings.

Variable				*N*	Mean *ES*	Difference (<.05)
Year			2013–2017	11	0,898	Yes
			2018–2023	10	0,846	
			2003–2007	10	0,391	
			2008–2012	6	0,171	
Sample (Country)			Malaysia	2	2,006	Yes
			Oman	1	1,992	
			Egypt	1	1,426	
			The Netherlands	1	1,210	
			Germany	2	1,054	
			South Korea	2	0,955	
			China	1	0,929	
			Turkey	1	0,872	
			Spain	3	0,795	
			Brazil	1	0,723	
			Taiwan	3	0,561	
			New Zealand	1	0,497	
			the USA	15	0,345	
			Italy	1	0,093	
			Multinational	1	−0,239	
			Mexico	1	−0,291	
Thinking Skill			Creative Thinking Skills	2	1,169	No
			Problem-solving skills	33	0,609	
			Critical Thinking Skills	2	0,513	
Education Level			University	20	0,690	No
			Elementary School	6	0,663	
			High School	5	0,655	
			Middle School	6	0,309	
AI Branch			Expert Opinion	3	0,506	No
			Intelligent tutors or agents	28	0,692	
			Personalised learning systems or environments (PLS/E)	6	0,400	
AI Tool or Agent	A	T1	Learning analytics through Moodle	1	0,872	Yes
	A	T1	PAT2Math	1	0,723	
	A	T1	U-LMS	1	−0,291	
	B	T1	Flowchart-based ITS (FITS)	2	2,006	
	B	T2	Educational robotics	1	1,992	
	B	T1	The Math Story Problem Tutor (MAST): An ITS	1	1,426	
	B	T1	SITS-Solution-based Intelligent tutoring system	1	1,388	
	B	T1	A Computer Program: An ITS	1	1,210	
	B	T1	PGBM-COMPS: An ITS	1	1,135	
	B	T1	Alexa (An SPA-Smart Personal Assistant): An ITS	2	1,056	
	B	T1	Graphical User Interface (GUI)	2	0,982	
	B	T1	iTutor: A problem solving oriented ITS	1	0,929	
	B	T1	ITS Hypergraph Based Problem Solver (HBPS)	1	0,535	
	B	T1	Andes: A mature ITS	5	0,523	
	B	T1	SQLT-Web: An ITS	1	0,497	
	B	T1	HelpNeed and Use: An ITS	1	0,450	
	B	T1	Model Tracing Intelligent Tutor (MIT)	1	0,403	
	B	T1	ProPL-a dialogue-based tutoring system	1	0,402	
	B	T1	Logiocando: A hypermedia ITS	1	0,093	
	B	T1	Transaction Analysis Tutor: An ITS	1	−0,040	
	B	T1	Animal Watch: An ITS	3	−0,095	
	B	T1	Computer-based assessment	1	−0,239	
	C	T3	STEM-based AI Program	1	0,776	
	C	T1	The interactive e-book learning system (SimMagic)	1	0,735	
	C	T2	Personal Audio Classifier (PAC)	1	0,610	
	C	T3	Online Platform for Collaborative Case Study	1	0,213	
	C	T1	Simple Machines Learning Environment (SIMALE)	2	−0,098	

Note: A = Expert Opinion, B = Intelligent tutors or agents, C = Personalised learning systems or environments (PLS/E), T1 = Problem-solving skills, T2 = Creative thinking skills, T3 = Critical thinking skills

of AIEd (Brown et al., 2020), and additionally, the growing rate at which AIEd tools influences HOT skills is discussed elsewhere (e.g. Spector & Ma, 2019; Xia & Li, 2022).

This study revealed that the magnitude of effect sizes varied significantly across countries. In other words, the country (or sample) variable moderated the magnitude of effect sizes significantly. Consequently, differences among nations in their AIEd practices could be observed clearly (see Table 3). Accordingly, the studies in Malaysia have the largest mean effect size and the United States has the greatest number of research ($N = 15$). This finding of the U.S. outnumbering other countries in their AI studies in educational contexts has been reported in several research (Chen et al., 2020; Hopcan et al., 2023). Hopcan et al. (2023) also pointed to worldwide interest in the use of AIEd. While it's quite reassuring to find that AI use is widespread in the world, since far majority of the samples in our meta-analysis have one or two studies completed, more research on this topic needs to be undertaken to have a better understanding of potential changes in AIEd tools' influence on HOT skills in different countries.

The results confirmed the nonsignificant difference in effect sizes among different types of HOT skills, namely critical thinking, creative thinking, and problem-solving. The extensive application and exploration of technologies such as AI, VR/AR, and 5G is found to increase creativity, enhance effective learning, and provide a productive and positive learning environment (Hew & Kadir, 2016; Wang et al., 2021). Schools today are better equipped with different sorts of technologies, which suggest effective practice conditions in the classroom that address each student's learning strengths and weaknesses (Lin & Chang, 2020; Muthmainnah et al., 2022). For example, during the learning process, AIEd employs language analysis and speech recognition techniques to evaluate how students think and behave (Vincent-Lancrin & Van der Vlies, 2020). With so many support elements and promising outcomes in terms of HOT skills, AI technologies serve enhancement of all those thinking skills equally well. According to Table 3, problem-solving skill ($N = 33$) was the most frequently tested HOT skills. The relatively higher number of studies completed on AI's role in the enhancement of problem-solving skills could be explained by the growing globalisation and interconnectedness in nations. To elaborate further, the challenges faced are becoming increasingly intricate and intractable due to this phenomenon. Consequently, there is a rising demand for skills to tackle new challenges and a sophisticated comprehension of how to confront them effectively (Joksimovic et al., 2023). This might have likely led to a predominant focus on research employing AI for the enhancement of problem-solving abilities. Taken together, keeping potential contribution of AI technologies to the development of HOT skills (Sakulkueakulsuk et al., 2018; Spector & Ma, 2019; Zhan et al., 2024) and the relatively small number of studies completed for creative and critical thinking skills, more studies are needed to ensure better understanding of the influence of AI technologies on different types of HOT skills.

The classification of AI tools/ agents as "AI branches" to see their impact on the development of HOT skills revealed nonexistence of significant difference. In other words, AI tools/agents moderated the development of HOT skills at similar magnitudes when they were used to assist expert opinion, act as intelligent tutors or agents or provide personalised learning systems or environments. Additionally, summary Table 3 revealed the fact that researchers who had investigated the impact of their distinct AI tools on problem-solving utilised those tools either to assist expert opinion or serve as intelligent tutors. Effects on critical thinking skills were examined through the use of AI tools as personalised learning systems or environments. On the other hand, distinct AI tools/ agents were found to have significantly different effects on HOT skills, with "Flowchart-based ITS (FITS)" having the highest impact (ES = 2,006) and "U-LMS" the lowest (ES = −0,291). The review of findings on distinct use of AI tools/agents in different contexts informs inconclusive results regarding their effects on various HOT skills. For example, in two quasi-experimental studies (Hooshyar et al., 2015; Hooshyar et al., 2016) examining a novel FITS aimed to enhance students' problem-solving skills in computer programming, considerable gains were observed for the experimental group over the control group. On the contrary, in their study examining the effect of a "ubiquitous learning management system (U-LMS)", a system composed of programmes and a content repository, which would "facilitate content authoring, sequencing, tracking students – system interaction, and

assessment" (p. 304) of domain knowledge and self-regulated learning content, Cárdenas-Robledo and Peña-Ayala (2019) found better results for the students in control group, who were instructed traditionally by professors. One of the explanations they suggested was that the experimental group had another stimulus (U-LMS) which increased their workload. Similarly, while *Logiocando*, a hypermedia intelligent tutor designed to help primary school students (9–10 years old) to learn the basic concepts of logic (Lanzilotti & Roselli, 2007) was found to have low level influence on problem-solving (ES = 0,093), the effect of SITS, an intelligent tutoring system to enhance problem-solving skills in computer programming (Hooshyar et al., 2018) was at a high level (ES = 1,388). Considering the mixed results obtained from limited number of extant research examining the impact of distinct AI tools/agents on various HOT skills in different contexts, further work needs to be done to enhance our understanding of effective integration of AI technologies in educational settings. AI technologies are yet to be fully investigated not only in terms of their potential effects on development of HOT skills, but also their affordances and limitations including several factors in instructional settings, such as their integration to existing curricula, their design features, students' readiness level or acceptance of such technologies, or teaching roles (Pence, 2019; Xia & Li, 2022) should be explored.

Technology has an integral role in our modern world now, where it has found many uses in different disciplines and fields (Zhuang, 2022). The incorporation of technological advances into educational practices is inevitable, and notably the use of technology for HOT development is receiving greater significance. Consequently, over the past two decades, researchers have grown interest towards AI technologies, which may also be the result of support and guidance provided by international organisations such as the World Economic Forum, OECD, and the National Research Council (Hopcan et al., 2023). These organisations have acknowledged that, similar to the significance given to HOT skills, technological concepts and technological literacy are crucial skills for the twenty-first-century (ISTE, 2021; National Research Council [NRC], 2012). Together with a number of reviews on AIEd (Chen et al., 2020; Drigas & Ioannidou, 2012; Fahimirad & Kotamjani, 2018; Hwang & Tu, 2021; Zawacki-Richter et al., 2019) in literature, this study contributes to the expansion of understanding the impact AI technologies have on HOT skills. This meta-analysis identified medium positive influence of AI technologies on students' HOT skills, and when incorporated into classroom settings or practices, they could serve enhancement of different HOT skills equally well, in all education levels and nations worldwide, with a growing trend of impact each year. The implications that would follow can be exemplified as providing more support and assistance towards integration of AI technologies to educational settings (Xia & Li, 2022) and increasing educators' knowledge and skills in successful application of those technologies in their classes.

Limitations and suggestions for future work

This study reviews the literature on the effect of AIEd on HOT skills and draws out a broad picture of the current situation in the field. The findings might lay the groundwork for the execution of AIEd in the future; however, this study is constrained by several limitations. Firstly, mean effect size calculations and moderator variable examinations were performed with the inclusion of 29 articles that met the criteria among 2.440 studies. Besides, to ensure a level of credibility, only published articles were included; books, dissertations, and conference proceedings were excluded. Inclusion of additional studies in the future would expand the understanding of the effects of AIEd tools/agents. Secondly, the present study analysed the effects of AIEd tools/agents on HOT skills. Future studies could expand the research by including publications of other dependent variables (e.g. design capacities of instructors, learners' skills on using or adapting to AI technologies).

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Data availability statement

The data that support the findings of this study will be openly available in Mendeley.

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